ConfiGuard-Lite: Project Overview

As of 2026-11-10.

1. Overview

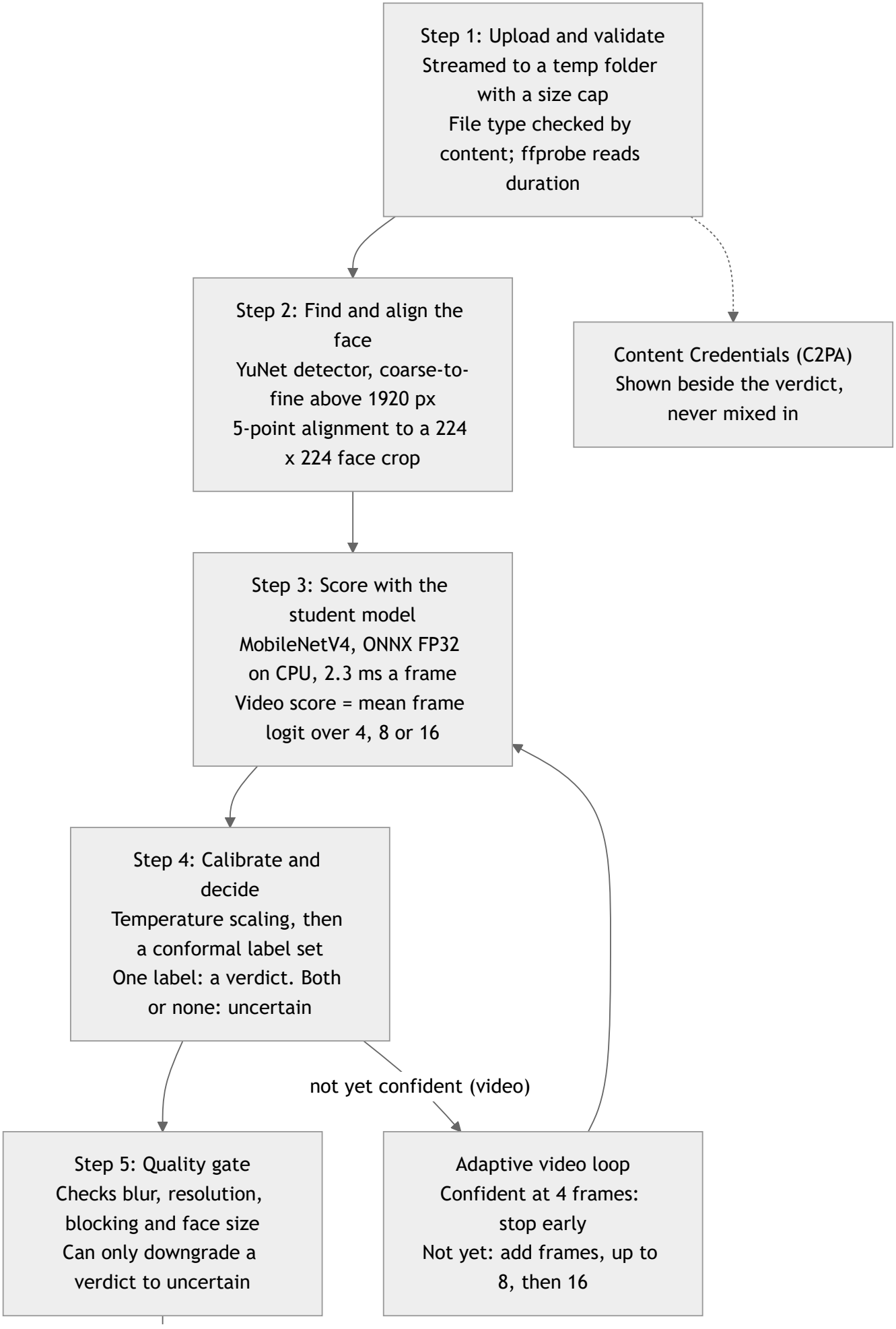
ConfiGuard-Lite is a deepfake detector for face images and videos. It is small enough to train on a 6 GB laptop GPU and to run on a CPU-only server. It answers with one of three verdicts: **likely real**, **likely manipulated**, or **uncertain**. It says "uncertain" rather than guess when the evidence is weak.

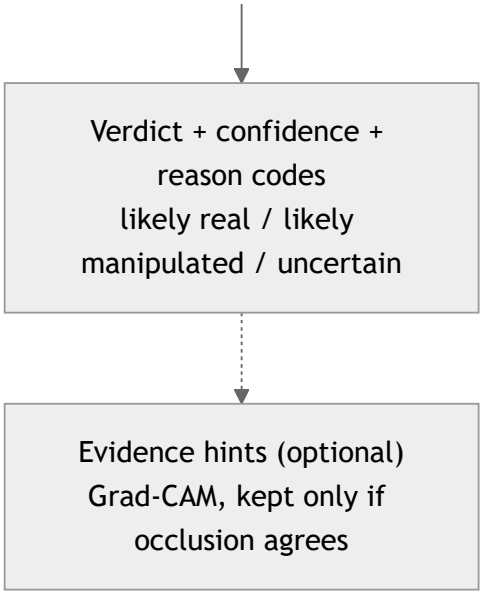
The core is a 2.49M-parameter MobileNetV4 model (9.5 MiB as ONNX). It learned from a 304M-parameter CLIP "teacher" through knowledge distillation. On FaceForensics++ validation videos it scores **0.974 video AUROC**. A video verdict takes about 6 frames on average, and the model itself runs in **8.8 ms** on CPU.

- **Built for:** journalists, moderators and researchers who need a quick, honest second opinion, not a forensic certificate.
- **Hardware target:** Windows laptop, Ryzen 7000 CPU, NVIDIA RTX 4050 Laptop GPU (6 GB VRAM). It deploys on CPU, with optional NVIDIA GPU.
- **What sets it apart:** it is uncertainty-aware, it degrades safely on low-quality media, and it reports Content Credentials (C2PA) provenance as a separate signal, never mixed into the score.
- **Scope:** results are validated on FaceForensics++ c23 development data only. The test split is still sealed, and every answer carries a "not proof" notice.

2. How it works end to end

An upload goes through five stages. Every stage after scoring can only make the answer more cautious. Images run each stage once on a single face crop. Videos sample 16 evenly spaced frame positions but score only as many as needed.


Step 1: Upload and validate
Streamed to a temp folder with a size cap
File type checked by content; ffprobe reads duration
Step 2: Find and align the face
YuNet detector, coarse-to-fine above 1920 px
5-point alignment to a 224 x 224 face crop
Content Credentials (C2PA)
Shown beside the verdict, never mixed in
Step 3: Score with the student model
MobileNetV4, ONNX FP32 on CPU, 2.3 ms a frame
Video score = mean frame logit over 4, 8 or 16
Step 4: Calibrate and decide
Temperature scaling, then a conformal label set
One label: a verdict. Both or none: uncertain
not yet confident (video)
Step 5: Quality gate
Checks blur, resolution, blocking and face size
Can only downgrade a verdict to uncertain
Adaptive video loop
Confident at 4 frames: stop early
Not yet: add frames, up to 8, then 16



Solid arrows are the decision path. Dashed arrows are side outputs that never change the verdict. A typical FF++ video is decided in about 0.5–1 s end to end.

Plain-text version of the same flow:

```
Upload -> Validate -> Face detect + align -> Student model (ONNX)
       -> Calibration + conformal set -> Quality gate -> Verdict
                ^                 |
                +-- add frames ---+   (video only: 4 -> 8 -> 16 frames)

Side outputs (never change the verdict): C2PA provenance, evidence heatmaps
```

3. Tech stack and why each piece was chosen

Every choice serves two limits: training must fit 6 GB of VRAM, and serving must run on a plain CPU server. So PyTorch is used for training only. The deployed service runs ONNX Runtime and never imports PyTorch.

Layer	Technology	What it does here	Why this one
Language	Python 3.11	All ML, data and service code	The ML stack (PyTorch, timm, ONNX Runtime) supports it fully. 3.14 is not yet supported.
Training	PyTorch 2.5.1 + CUDA 12.1, fp16 AMP	Trains the student on the RTX 4050	Mixed precision and gradient accumulation keep peak VRAM tiny: 154 MB at batch 8.
Model zoo	timm	Supplies the MobileNetV4-Conv-Small and EfficientNet-B0 backbones	Pinned, ImageNet-pretrained weights with a uniform API, so the two models can be swapped and compared fairly.
Teacher	Hugging Face Transformers + GenD CLIP-L/14	Frozen teacher that produces soft labels once, offline	Its forgery knowledge transfers to a model 120x smaller. It is never trained here, so it fits the laptop.

Layer	Technology	What it does here	Why this one
Face detection	OpenCV YuNet (ONNX)	Finds faces and 5 landmarks per frame	Fast on CPU and needs no PyTorch. The model file is pinned by SHA-256.
Media	OpenCV, FFmpeg / ffprobe	Decodes video frames, reads duration and frame counts	ffprobe gives reliable metadata for variable-frame-rate and rotated video. OpenCV decodes the pixels.
Inference runtime	ONNX Runtime (CPU, optional CUDA)	Runs the exported student in production	2.3 ms per frame on CPU, no PyTorch on the server, and the same file serves CPU and GPU.
Model formats	ONNX FP32 / FP16 / INT8	Deployment variants	FP32 (9.5 MiB) is the default. FP16 is equally accurate but not faster here. INT8 is not used because its agreement was lower.
Statistics	NumPy	Temperature scaling, conformal prediction, quality signals	Small, auditable math with no extra dependencies.
API	FastAPI + Uvicorn	`/v1/analyze`, health checks, OpenAPI docs	Async streaming uploads, typed schemas and automatic API docs.
Provenance	c2pa-python (official CAI SDK)	Reads and verifies Content Credentials	It is the reference implementation, checked against the official C2PA Trust List.
Config	YAML + dataclasses, `.env`	Per-environment settings and paths	Fewer dependencies than pydantic. Machine paths and secrets stay out of the code.
Frontend	React 19, TypeScript, Vite, Tailwind CSS 4	Landing page, detector UI and about page	A typed, fast build. Its output is served by the same FastAPI process, so there is no CORS.
3D hero	three.js + a CC0 MakeHuman head	Animated landing visual	Loaded lazily, with a poster image fallback when WebGL is unavailable.
Quality tools	pytest, Vitest, Testing Library, Playwright, axe-core, Lighthouse, Biome	715 Python tests, 82 unit tests, end-to-end, accessibility and performance checks	Each layer is tested where it lives.
Packaging	Docker (multi-stage), Compose	Hardened image `configuard-lite:1.0.0-rc.1`	Runs as a non-root user with a read-only filesystem, all Linux capabilities dropped and a health check.
Hosting	Oracle Cloud Always Free (Ampere arm64) + Caddy	Public HTTPS deployment	Free with enough RAM (about 1 GiB peak). Caddy handles automatic HTTPS and caps upload size.

4. Models and training

The shipped model is `student_distilled_p80`. It is a MobileNetV4-Conv-Small student that learned from a frozen GenD CLIP-L/14 teacher on FaceForensics++ c23 face crops. It reaches about the teacher's video AUROC (0.974 vs 0.979) at 1/120th of the size.

Model	Role	Parameters	Val video AUROC	Notes
GenD CLIP-L/14	Frozen offline teacher	303.97M	0.979	MIT license. Only its logits are cached, and it never ships.
MobileNetV4-Conv-Small, distilled	**Production student**	2.49M	0.974 (calibrated)	Frame ECE 0.033 vs 0.071 for the baseline, so its probabilities are more honest.
MobileNetV4-Conv-Small, BCE only	Training baseline	2.49M	0.981	Slightly higher AUROC but badly calibrated (frame NLL 0.504 vs 0.216).
EfficientNet-B0	Comparison baseline	4.01M	—	About 2x slower on CPU (99.8 vs 46.7 ms P50 for the backbone).

Data. The model was trained on FaceForensics++ c23: 5,000 videos (8.4 GiB) on drive D:. They cover four manipulation methods: Deepfakes, Face2Face, FaceSwap and NeuralTextures. The authors' official split is used (720/140/140 original videos). From these, 79,280 aligned 224×224 face crops were extracted, 16 per video, with real and fake crops matched by frame index. Families that leak across splits were quarantined, and the test split is sealed.

Distillation. The loss is `(1−α)·BCE + α·T²·BCE(σ(z/T), σ(m/T))` with α = 0.5 and T = 2. Here m is the teacher's margin (logit_fake − logit_real). The student copies the teacher's confidence as well as its answers, which is where the better calibration comes from.

Training on 6 GB. Training uses fp16 mixed precision, AdamW with a warm-up cosine schedule, gradient clipping, and cached face crops so no decoding happens during training. Seeds are deterministic, and training can resume from any epoch.

Experiments that were tried and rejected, each with a recorded decision:

- **Compression-robust training (6e):** the worst-case stress AUROC rose from 0.692 to 0.825, but clean AUROC fell to 0.927, so this model was not shipped.
- **GRU temporal head (7):** a 157k-parameter GRU over frame embeddings gained +0.0014 AUROC, a confidence interval that includes zero. Video scoring stays as the mean frame logit.
- **INT8 export (8):** 2.7 MiB, but uncalibrated and with lower agreement, so it is not used.

5. The trust layer: knowing when not to answer

A raw score is not a verdict. Four layers sit around the model. Each one can only make the answer more cautious; none of them can flip real to fake.

1. **Calibration (temperature scaling).** This rescales the logit so that a stated 90% means right about 90% of the time. Video ECE dropped from 0.043 to 0.035. Single images and videos are calibrated separately.
2. **Conformal prediction gives the "uncertain" verdict.** Mondrian split conformal (α = 0.05) builds a set of plausible labels. One label gives a verdict. Both labels, or none, gives **uncertain** with a reason code. On validation, 5.8% of videos were uncertain, only 2.2% of real videos were called manipulated, and 92.2% of decided videos were correct.
3. **Media-quality gate.** Sharpness, effective resolution, blocking and face size are measured on every crop. Poor quality turns a confident verdict into uncertain, for example `LOW_SHARPNESS` or `SMALL_FACE`.

On heavily blurred real videos (blur σ 2), false accusations fell from 84% to 0%.

4. **Evidence hints (explanations).** Optional Grad-CAM heatmaps on the face crop show where the score came from. A hint is shown only if an occlusion check confirms it; otherwise an occlusion map is tried, and if that also fails, nothing is shown. About half are withheld. Hints are labelled "not proof" and never change the verdict.

Content Credentials (C2PA) is a separate signal. If a file carries a C2PA manifest, the official CAI SDK verifies it against the official C2PA Trust List. It reports who signed the file and what they declared. This status is shown next to the model's verdict, never merged into it. Most files have no credentials, and that says nothing about manipulation.

6. Service, website and deployment

One FastAPI process serves both the API and the React website. It runs the ONNX model on CPU and ships as a hardened Docker image.

API (`src/configuard/service/`)

- `POST /v1/analyze` takes one image or video. `GET /v1/limits` returns upload limits. `GET /health/live` and `GET /health/ready` report health. OpenAPI docs are served at `/docs`.
- **Upload safety:** uploads are streamed to a per-request temporary folder with a byte cap. The file type is checked by its content (magic bytes) against its extension, and video duration is checked with ffprobe. The temporary files are deleted after every request.
- **Load control:** a bounded worker pool and queue return `503 server_busy` instead of falling over. A per-request deadline returns `504`.
- **Integrity:** readiness requires every model artifact to match its SHA-256, re-checked every 30 s. Production refuses to start without API keys.
- **Privacy:** logs are structured JSON with request IDs and never contain file names. Uploads are never kept.
- **Limits (dev):** images up to 20 MB, videos up to 200 MB and 120 s. Accepted types are .jpg .jpeg .png .webp .mp4 .mov .mkv .avi.

Website (`frontend/`)

- **Pages:** `/` is the landing page, `/detect` the detector, and `/about` covers the method, evaluation and limits.
- **Built with:** React 19 and TypeScript, built by Vite, styled with Tailwind 4 and self-hosted fonts. It loads nothing from other domains, and a strict Content Security Policy blocks inline scripts.
- **Detector page:** shows upload progress, the verdict with its confidence, the evidence timeline for videos, heatmap hints and a separate Content Credentials card.
- **Accessibility and themes:** light, dark and system themes, reduced-motion support, and axe-core accessibility tests.

Deployment

- **Docker image `1.0.0-rc.1`:** multi-stage build with hash-locked Python dependencies and digest-pinned base images. It runs as uid 10001 with a read-only filesystem, all capabilities dropped, and writes only to `/tmp`. A release manifest records the SHA-256 of every file.

- **Hosting (Phase 15):** Oracle Cloud Always Free (Ampere A1, arm64) behind Caddy, which provides automatic HTTPS, HSTS and a 160 MB body cap. The bundle and arm64 image were verified locally; the VM has not been created yet.

Running it locally

```
# Backend (from the project root, venv activated)
python scripts\serve.py --env development      # http://127.0.0.1:8000

# Frontend with hot reload (optional, second terminal)
cd frontend
npm ci                                          # first time only
npm run dev                                     # http://127.0.0.1:5173
```

7. How it differs from other deepfake detectors

Most deepfake projects chase a higher accuracy number with a bigger model and always output a fake-or-real label. ConfiGuard-Lite instead aims for near-teacher accuracy at laptop scale, admits when it does not know, and ships as a hardened service. The comparison below describes typical project types from general knowledge, not measured rivals.

Aspect	Typical student / Kaggle project	Large research model (CLIP or ViT based)	ConfiGuard-Lite
Model size	Xception or EfficientNet, 20–80M params	300M+ params, needs a big GPU	2.49M params, 9.5 MiB ONNX
Runs on	GPU notebook	Data-centre GPU	Plain CPU server; GPU optional
Output	Fake or real, always	A score	Likely real, likely manipulated, or **uncertain** with a reason
Probabilities	Uncalibrated	Rarely calibrated	Temperature-scaled + conformal sets
Bad-quality input	Confidently wrong	Often confidently wrong	Quality gate downgrades to uncertain
Video	Every frame, or a fixed N	Fixed clip	Adaptive 4→8→16 frames, about 6 on average
Explanations	Raw Grad-CAM, if any	Varies	Hints shown only after a faithfulness check, labelled "not proof"
Provenance	None	None	C2PA Content Credentials, kept separate from the score
Engineering	Notebook	Research code	Typed modules, 715 tests, SHA-pinned artifacts, hardened Docker
Evaluation honesty	Often train/test leakage	Varies	Official split, leakage audit, sealed test set, rejected experiments recorded

What stands out most:

- **Distillation from a frozen giant.** It borrows a 304M-parameter teacher's knowledge without ever training that teacher on a laptop.
- **Abstention as a feature.** "Uncertain" is a designed outcome with a statistical basis, not a failure.
- **Cost-aware video.** The analysis stops early when it is confident, cutting frames read by 61%.
- **Safe under bad input.** Blur, low resolution and tiny faces push the answer toward "uncertain", never toward an accusation.
- **Two separate questions.** "Does this look manipulated?" (the model) and "Who signed this file?" (C2PA) are reported side by side and never blended.

8. Results, limitations and what's next

On FaceForensics++ validation (695 videos), the production pipeline decides 86.8% of videos at 95.9% accuracy. It flags only 1.44% of real videos as manipulated and reads 6.2 frames per video on average. These are development numbers: the test split is sealed, and no other dataset has been evaluated.

Measure (FF++ c23 val, adaptive video)	Value
Video AUROC	0.973
Accuracy on decided videos	95.9%
Uncertain	13.2%
Real videos called manipulated (FPR)	1.44%
Fakes missed	4.1%
Frames read per video (average)	6.19, 61% fewer than 16
Model time per video on CPU (P50 / P95)	8.8 ms / 35.7 ms
End-to-end on a 4K 12 s clip	17–25 s (decoding dominates)

Known limitations

- **Weakest manipulation:** NeuralTextures. 29.5% of those videos come out uncertain and only 59% are detected outright.
- **Degraded media:** strong blur or downscaling can make real videos look fake. The quality gate catches much of this, but blur plus noise slips past it, and some low-resolution false accusations remain.
- **Generalisation unknown:** there are no results on other datasets, new generators (diffusion-made faces) or the sealed test split.
- **Faces only:** a video without a usable face, such as drone footage, comes out "uncertain". Fully synthetic scenes and voice cloning are out of scope.
- **Not evidence:** every answer is labelled an automated estimate, not a forensic determination.

What's next

- [ ] Create the Oracle Cloud VM and go live behind Caddy (Phase 15 is partial).
- [ ] Unseal the FF++ test split once, for a final honest number.
- [ ] Evaluate on other datasets (for example Celeb-DF, DFDC) and on diffusion-generated faces.
- [ ] Fix the shortcut where degraded real faces look fake, then recalibrate.
- [ ] Speed up 4K decoding.

Further reading in this repository

`README.md` (setup and usage), `docs/ARCHITECTURE.md` (module-level design), `docs/DECISIONS.md` (why each choice was made), `docs/MODEL_CARD.md` (model details), `docs/EXPERIMENT_LOG.md` (commands and results), `docs/KNOWN_ISSUES.md`, `docs/DATASETS.md`, `docs/DEPLOYMENT.md`.